Data-Driven Investigation of Culinary Patterns in CulinaryDB

Exploratory Analysis Inspired by Singh & Bagler (2015)

"Data-driven investigations of culinary patterns in traditional recipes across the world"

Name: Manthan Kapoor(2024336), Satyam Singh(2024514)
Professor: Dr. Ganesh Bagler
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This report summarizes the full exploratory analysis performed on the CulinaryDB dataset. All figures, tables, methodology notes, and findings were produced by running eda_culinarydb.ipynb.

1. Project Overview

Objective: Describe recipe and ingredient patterns across world cuisines using the CulinaryDB database. The work is inspired by Singh and Bagler's culinary pattern paper, but deliberately avoids analyses that require external flavor-molecule databases.

Scope

- Descriptive EDA only - no machine learning or prediction
- Paper-inspired plots: category heatmap, recipe-size distribution, rank-frequency curves (log-log and linear)
- Additional cuisine comparisons and top-ingredient summaries
- Honest documentation of analyses that cannot be reproduced without FlavorDB

Tools and libraries

- json, zipfile, io, pathlib - dataset loading and extraction
- pandas - DataFrame creation, grouping, aggregation, CSV export
- collections.Counter, defaultdict, sets - ingredient frequency and category counting
- numpy - reproducibility seed and numeric support
- matplotlib.pyplot - line plots, bar charts, figure export
- seaborn - heatmap for ingredient category composition

2. Dataset and Loading

Source: CulinaryDB relational repository (01_Recipe_Details.csv, 02_Ingredients.csv, 03_Compound_Ingredients.csv, 04_Recipe-Ingredients_Aliases.csv). The pipeline ingests normalized tables and aliases for all cuisine-labeled recipes.

Dataset summary (CulinaryDB corpus)

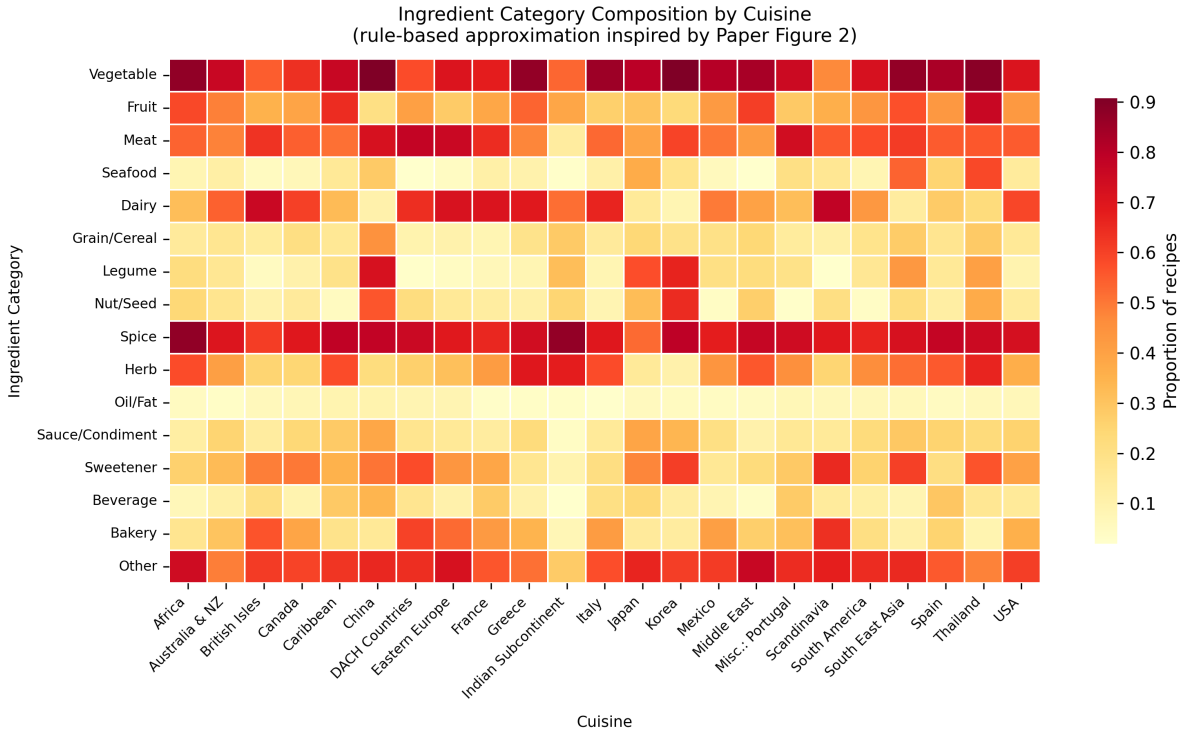
- Total recipes: 45,772
- Number of cuisines: 26
- Unique ingredients: 930 catalog entities (+ 103 compound blends)
- WORLD average recipe size: 8.44 ingredients
- Most represented cuisine: USA (16,118 recipes)
- Least represented cuisine: Misc.: Central America (14 recipes)

Data preparation steps

- Created recipe_size = count of distinct ingredients for each recipe
- Created ingredient_set per recipe so each ingredient counts once per recipe
- Checked missing values and duplicate IDs - no rows removed
- Built ingredient frequency table: number of recipes containing each ingredient

3. Visual Analysis - Figures and Insights

Figure 1 - Ingredient Category Composition by Cuisine



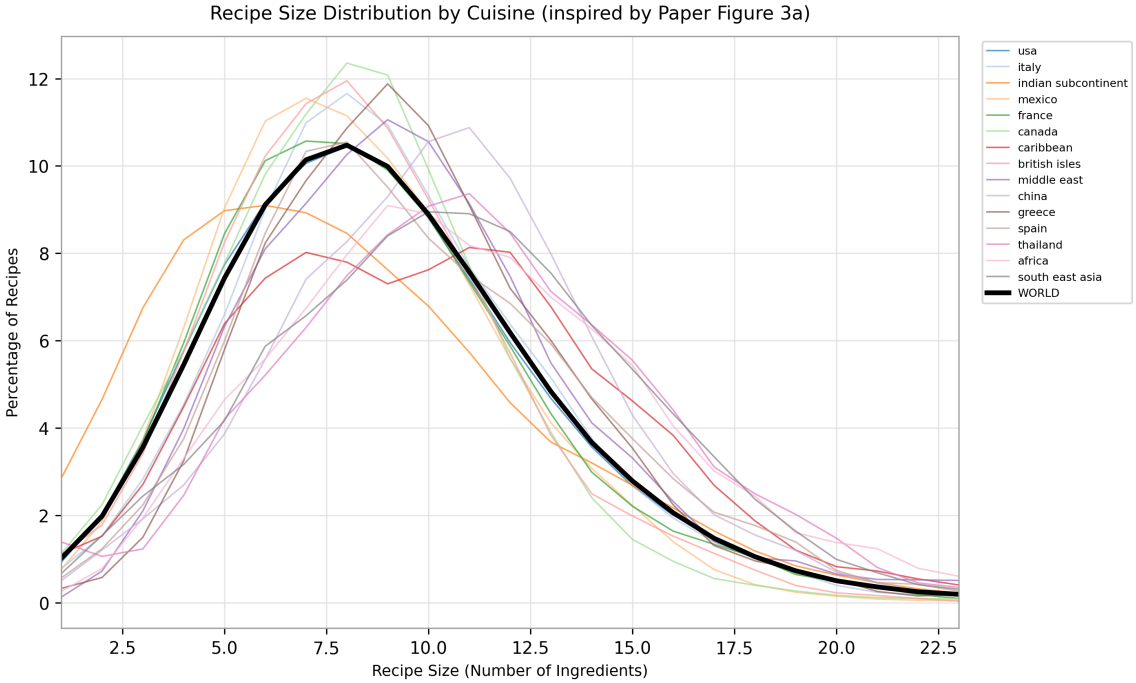
How this was created

Libraries: pandas, collections.defaultdict, seaborn, matplotlib. Each ingredient was mapped to one of 16 rule-based categories using keyword matching (assign_ingredient_category). For every recipe, we identified which categories appear at least once. For each cuisine, we divided the number of recipes containing a category by the total recipes in that cuisine to get a proportion between 0 and 1. The result was pivoted into a cuisine x category matrix and plotted with sns.heatmap (YlOrRd colormap).

Key insights

- Categories are an approximation - CulinaryDB aliases map to 16 high-level groups.
- Spice, Vegetable, and Sauce/Condiment categories appear in a large share of recipes across cuisines.
- Cuisine-specific differences are visible in the mix of Meat, Seafood, Dairy, and Legume categories.
- Normalization allows fair comparison despite uneven recipe counts per cuisine.

Figure 2 - Recipe Size Distribution by Cuisine (Paper Figure 3a inspired)



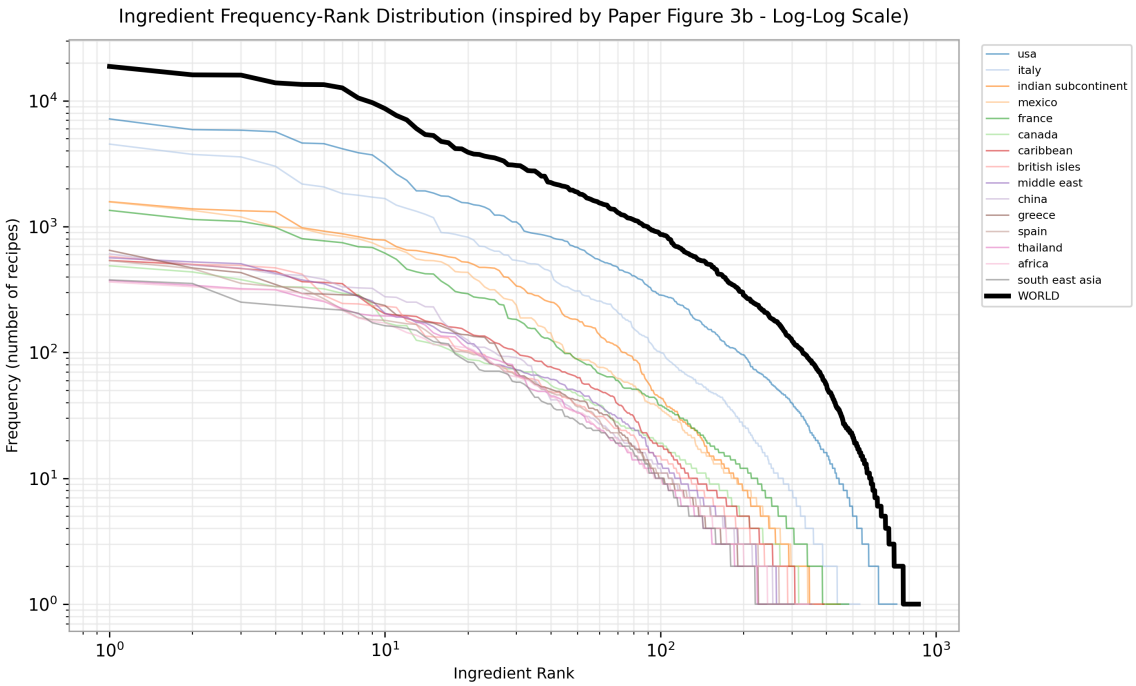
How this was created

Libraries: pandas, matplotlib. recipe_size was already stored as a column. For each cuisine and for WORLD (all recipes), we computed value_counts of recipe_size, converted counts to percentages, and plotted percentage vs. ingredient count as line plots. Individual cuisines use thin lines; WORLD uses a thick black line. X-axis limited to the 99th percentile for readability.

Key insights

- WORLD mean recipe size in this dataset is about 8.44 ingredients.
- The paper reports ~9 ingredients on average in their study across global repositories.
- Most cuisines show a unimodal distribution: recipe sizes rise to a peak then fall.
- Thai, African, and South East Asian cuisines tend toward larger average recipe sizes.

Figure 3 - Ingredient Frequency-Rank Distribution (Log-Log Scale, Paper Figure 3b inspired)



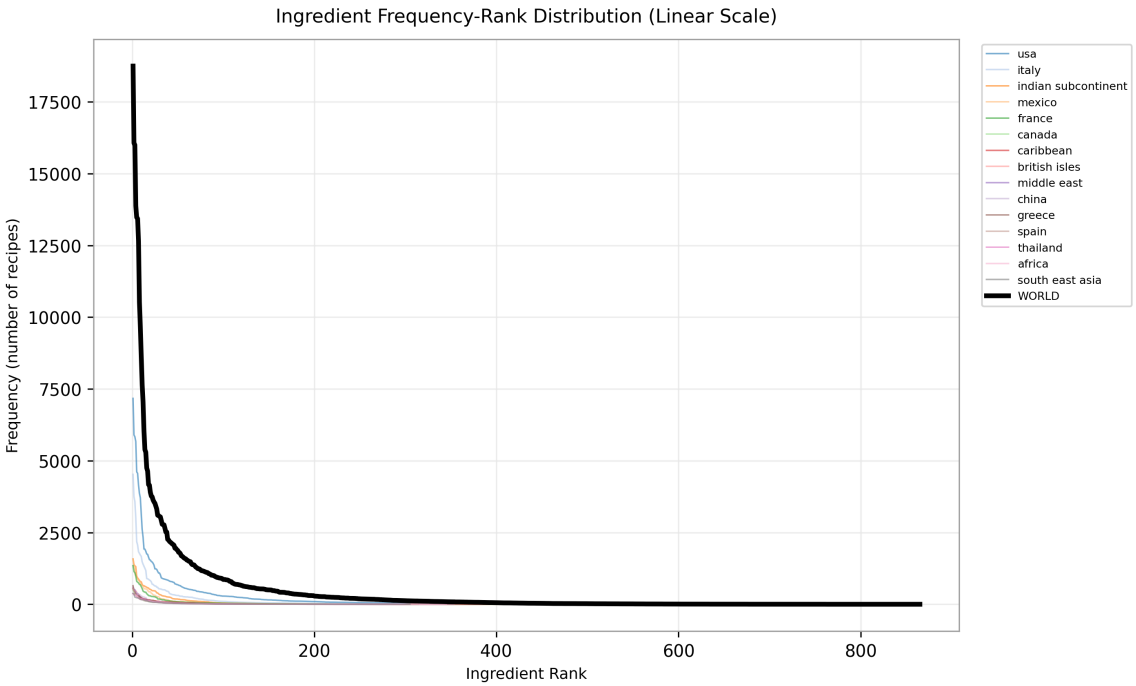
How this was created

Libraries: pandas, collections.Counter, matplotlib. For each cuisine and WORLD, we counted how many recipes contain each ingredient (using ingredient_set). Frequencies were sorted descending and assigned ranks 1, 2, 3, ... Rank vs. frequency was plotted on a log-log scale with thin cuisine lines and a prominent WORLD curve.

Key insights

- A steep drop from rank 1 to higher ranks appears in every cuisine.
- A few staple ingredients dominate; most ingredients appear in relatively few recipes.
- The WORLD curve summarizes the global head-heavy ingredient popularity pattern.
- Individual cuisines differ in how quickly frequency falls as rank increases.

Figure 3b - Ingredient Frequency-Rank Distribution (Linear Scale)



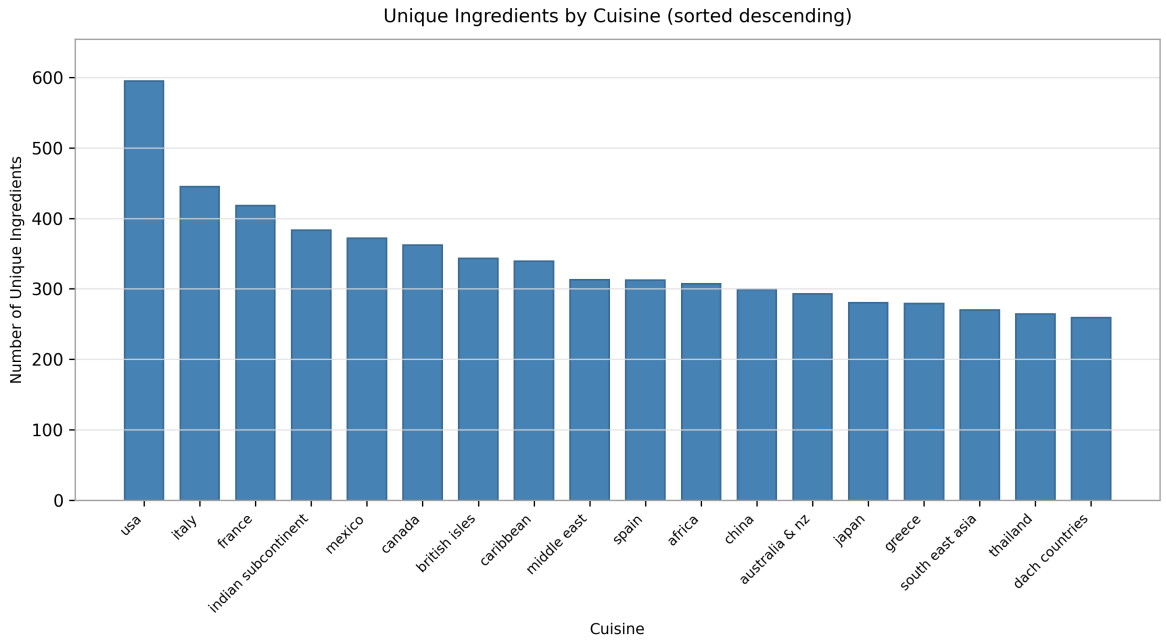
How this was created

Libraries: pandas, collections.Counter, matplotlib. For each cuisine and WORLD, we counted how many recipes contain each ingredient (using ingredient_set). Frequencies were sorted descending and assigned ranks 1, 2, 3, ... Rank vs. frequency was plotted on a linear scale with thin cuisine lines and a prominent WORLD curve.

Key insights

- The linear scale clearly reveals the extreme power-law disparity in ingredient usage.
- A tiny handful of primary staples (salt, garlic, onion, pepper) account for massive recipe counts.
- The frequency curve plunges rapidly and flattens out into an extensive long tail of specialized ingredients.
- Individual cuisines and the global WORLD profile follow the identical heavy-tailed pattern in linear space.

Figure 4 - Unique Ingredients by Cuisine



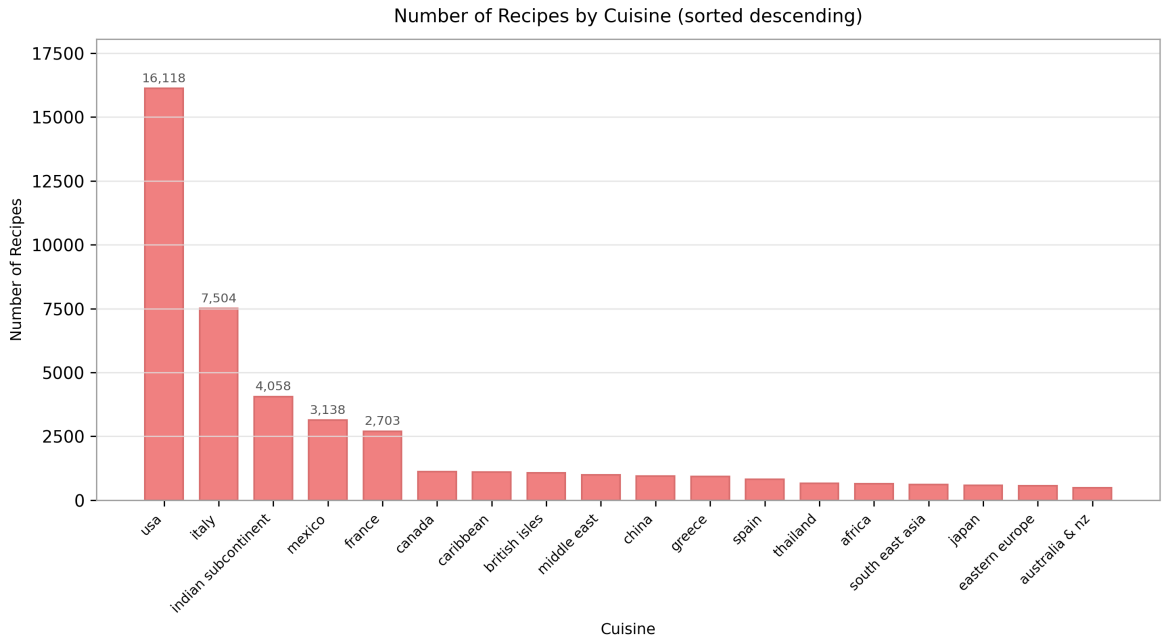
How this was created

Libraries: pandas, matplotlib. For each cuisine, we collected all ingredients from ingredient_set across recipes and counted distinct values with set union. Results were sorted descending and plotted as a bar chart.

Key insights

- USA has the largest ingredient vocabulary (595 unique entities in CulinaryDB).
- Italy (445) and France (418) also rank exceptionally high in unique ingredient count.
- Smaller cuisines like Scandinavian still show substantial ingredient diversity in the dataset.
- This measures breadth of ingredients present in the data, not real-world culinary completeness.

Figure 5 - Number of Recipes by Cuisine



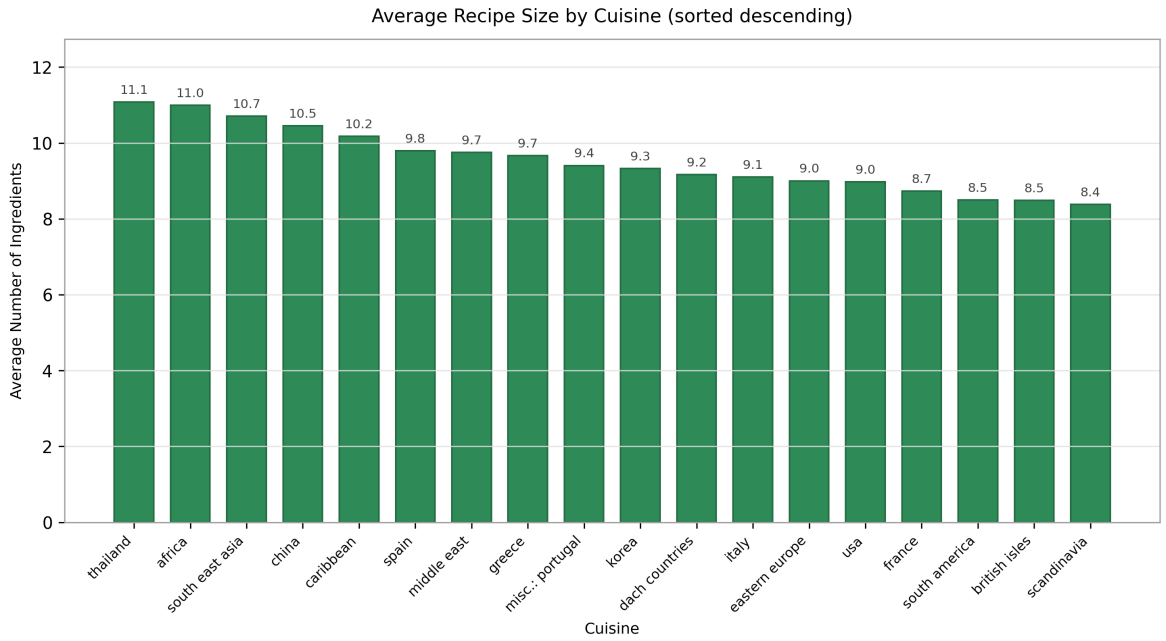
How this was created

Libraries: pandas, matplotlib. Used df['cuisine'].value_counts(), reset to a DataFrame, sorted descending, and plotted as a bar chart with optional count labels on taller bars.

Key insights

- USA has the most recipes in the corpus (16,118).
- Italy (7,504) and Indian Subcontinent (4,058) represent the next largest cohorts.
- The dataset is imbalanced across cuisine classes.
- This imbalance is noted for context only - no classification model was built.

Figure 6 - Average Recipe Size by Cuisine



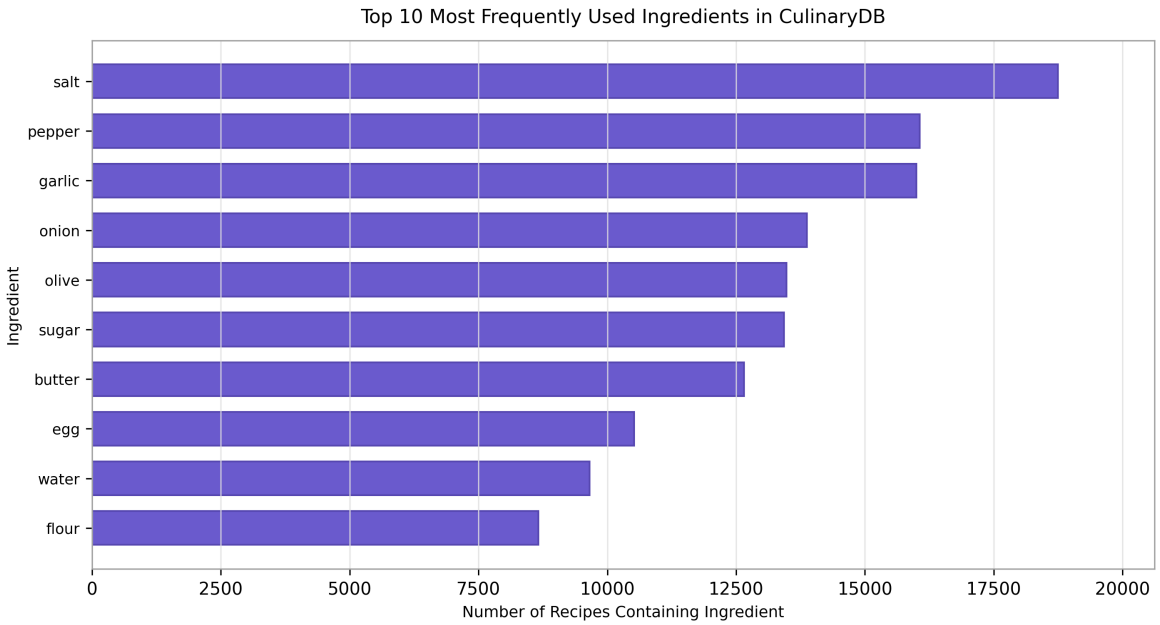
How this was created

Libraries: pandas, matplotlib. Grouped by cuisine and computed count, mean, median, min, and max of recipe_size. Sorted by mean descending and plotted mean recipe size as bars with value labels. Full table saved to recipe_size_statistics.csv.

Key insights

- Highest mean recipe size: Thailand (11.08 ingredients) and Africa (10.99 ingredients).
- Lowest mean recipe size: Canada (8.28 ingredients) and Australia (8.32 ingredients).
- Indian, Thai, and Caribbean recipes tend to list more ingredients on average.
- Average size is descriptive only - it does not rank cuisines by quality or complexity.

Figure 7 - Top 10 Most Frequent Ingredients Overall



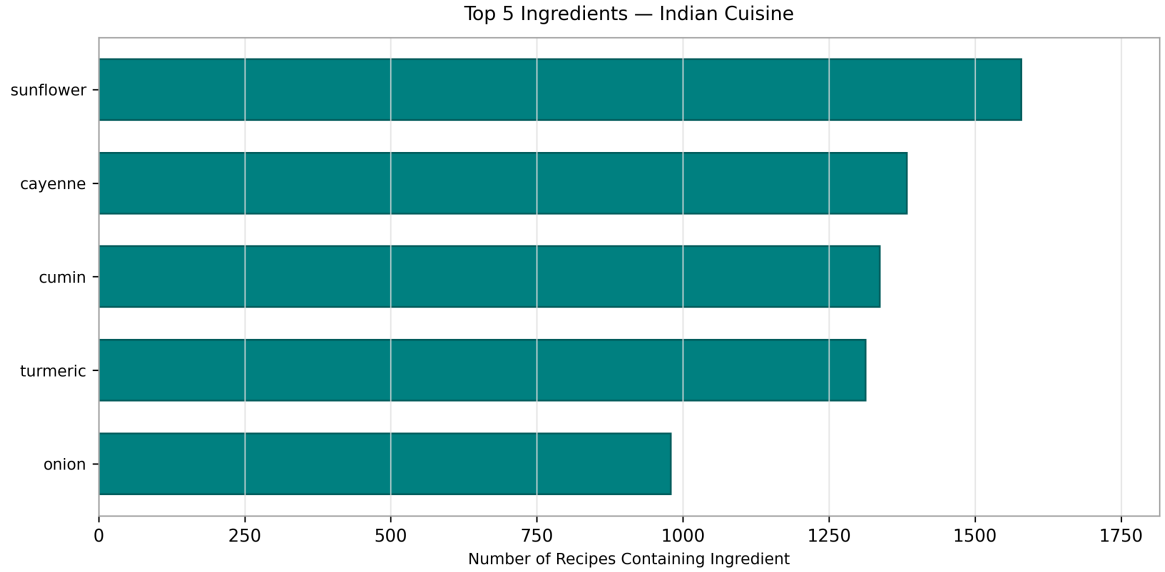
How this was created

Libraries: pandas, collections.Counter, matplotlib. Counted recipes containing each ingredient across the full training set (one count per recipe per ingredient). Selected top 10 by recipe_count and plotted a horizontal bar chart.

Key insights

- The most common ingredient is salt (18,745 recipes).
- Basic staples (salt, pepper, garlic, onion, olive oil) dominate the top ranks.
- Frequency means number of recipes containing the ingredient, not total string occurrences.
- These patterns reflect common recipe-writing conventions in the CulinaryDB collection.

Figure - Top 5 Ingredients in Indian Cuisine



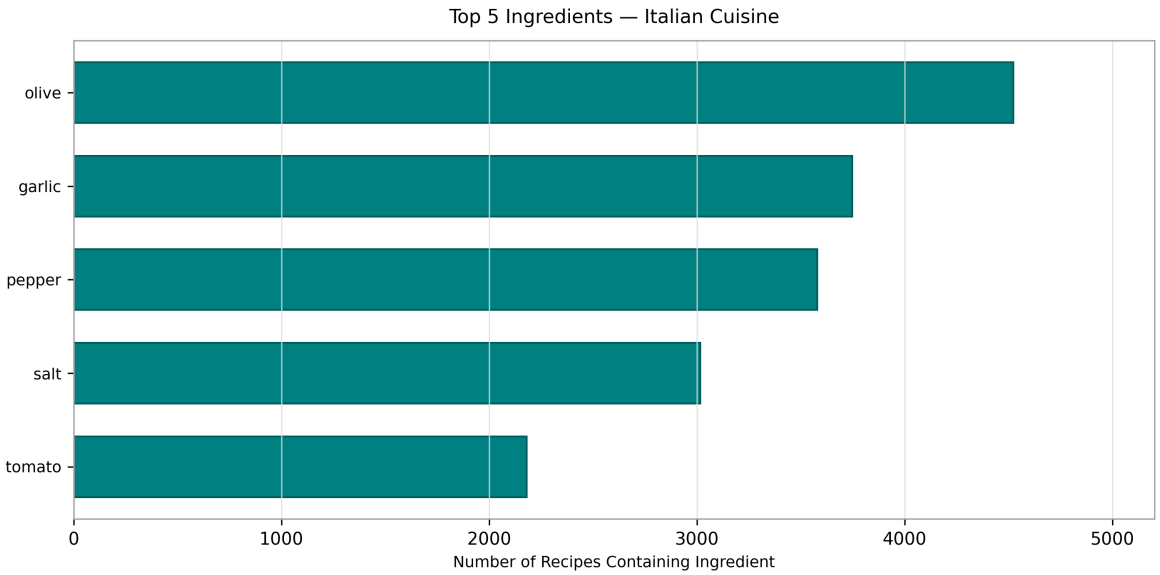
How this was created

Libraries: pandas, collections.Counter, matplotlib. Filtered df to one cuisine, counted recipes containing each ingredient using ingredient_set, selected top 5, and plotted horizontal bars. All four target cuisines (indian, italian, japanese, mexican) were present in the dataset.

Key insights

- Shows the five most common ingredients specifically within Indian recipes.
- Cuisine-level top ingredients differ from the global top 10.
- Spices, aromatics, and cuisine-specific staples (sunflower oil, cayenne, cumin, turmeric) dominate.
- Results describe this dataset subset, not every traditional recipe from that cuisine.

Figure - Top 5 Ingredients in Italian Cuisine



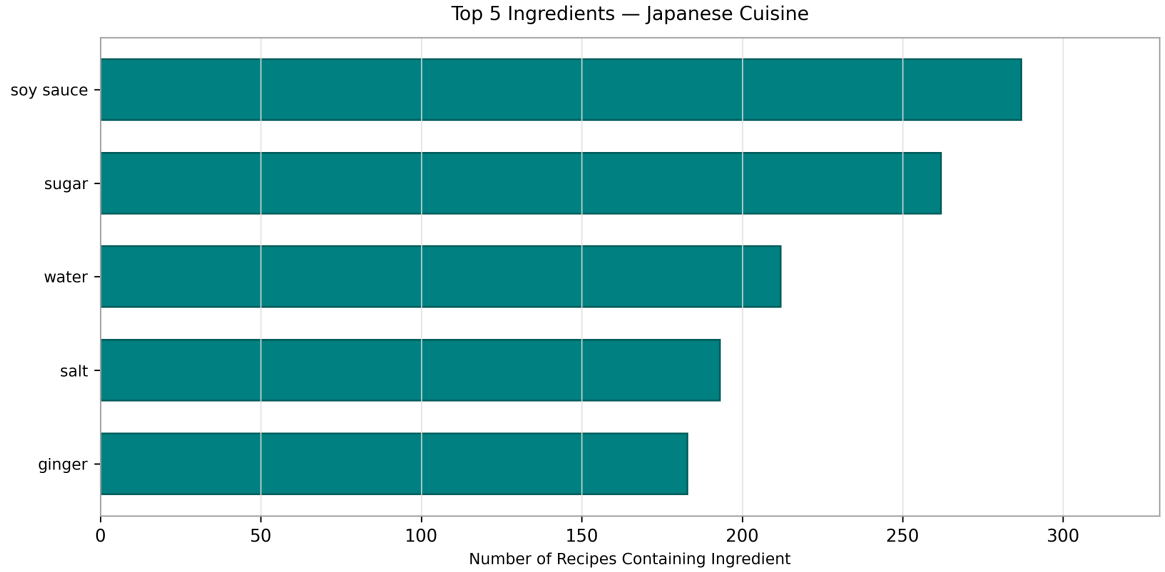
How this was created

Libraries: pandas, collections.Counter, matplotlib. Filtered df to one cuisine, counted recipes containing each ingredient using ingredient_set, selected top 5, and plotted horizontal bars. All four target cuisines (indian, italian, japanese, mexican) were present in the dataset.

Key insights

- Shows the five most common ingredients specifically within Italian recipes.
- Cuisine-level top ingredients differ from the global top 10.
- Olive oil, garlic, black pepper, and tomato establish the Mediterranean flavor profile.
- Results describe this dataset subset, not every traditional recipe from that cuisine.

Figure - Top 5 Ingredients in Japanese Cuisine



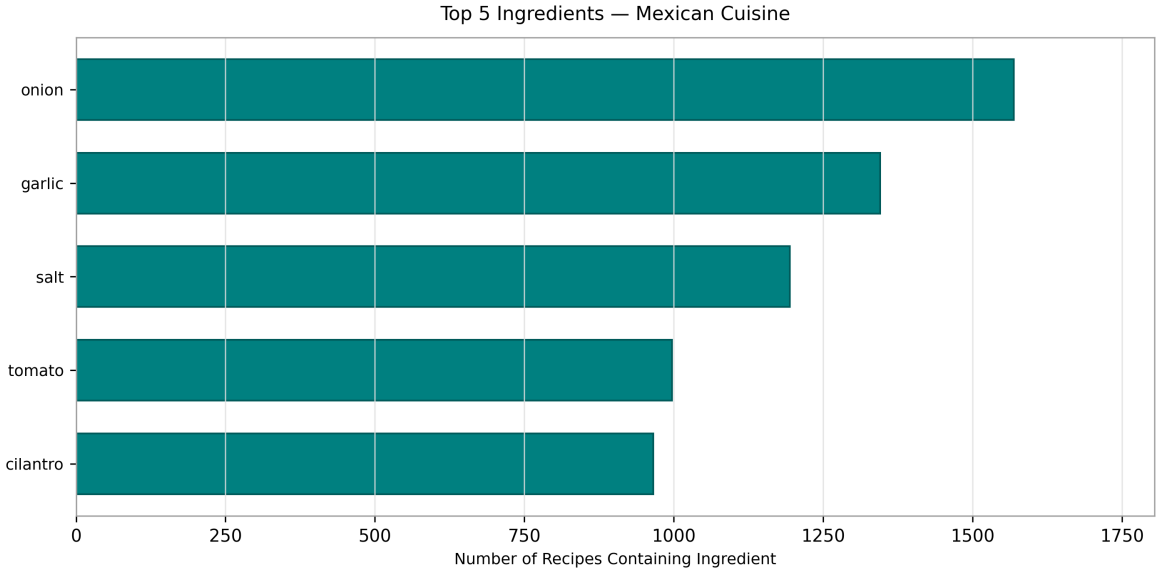
How this was created

Libraries: pandas, collections.Counter, matplotlib. Filtered df to one cuisine, counted recipes containing each ingredient using ingredient_set, selected top 5, and plotted horizontal bars. All four target cuisines (indian, italian, japanese, mexican) were present in the dataset.

Key insights

- Shows the five most common ingredients specifically within Japanese recipes.
- Cuisine-level top ingredients differ from the global top 10.
- Soy sauce, sugar, water, and ginger form the foundation of Japanese seasoning.
- Results describe this dataset subset, not every traditional recipe from that cuisine.

Figure - Top 5 Ingredients in Mexican Cuisine



How this was created

Libraries: pandas, collections.Counter, matplotlib. Filtered df to one cuisine, counted recipes containing each ingredient using ingredient_set, selected top 5, and plotted horizontal bars. All four target cuisines (indian, italian, japanese, mexican) were present in the dataset.

Key insights

- Shows the five most common ingredients specifically within Mexican recipes.
- Cuisine-level top ingredients differ from the global top 10.
- Onions, garlic, salt, tomato, and cilantro characterize Mexican flavor foundations.
- Results describe this dataset subset, not every traditional recipe from that cuisine.

4. Paper Analyses Not Reproducible with CulinaryDB Alone

Singh and Bagler's Figures 4 and 5 require mapping ingredients to flavor molecules via FlavorDB, computing food-pairing scores, randomized cuisine controls, and Z-scores. CulinaryDB provides id, cuisine, and ingredient mappings without volatile chemical compound profiles.

- No flavor molecules were invented or imported from external databases.
- No fake Z-scores or food-pairing scores were generated.
- No arbitrary similarity metric was substituted and labeled as the paper's result.
- This report and notebook focus on honest recipe/ingredient-level reproduction only.

5. Key Findings

- The training set contains 45,772 recipes across 26 cuisines.
- Average recipe size (WORLD) is 8.44 ingredients in CulinaryDB.
- Most frequent ingredient: salt (18,745 recipes).
- Largest cuisine by recipe count: USA (16,118), followed by Italy (7,504).
- Largest ingredient vocabulary: USA (595 unique ingredients).
- Recipe-size and rank-frequency curves show consistent head-heavy ingredient usage patterns.
- Category heatmap reveals cuisine-specific ingredient mix differences under rule-based mapping.

Saved outputs

- Figures: outputs/figures/ (PNG, 300 dpi)
- Tables: cuisine_summary.csv, ingredient_frequency.csv, unique_ingredients_by_cuisine.csv, recipe_size_statistics.csv, top_ingredients_overall.csv
- Notebook: eda_culinarydb.ipynb

6. References

- Singh, N., & Bagler, G. "Data-driven investigations of culinary patterns in traditional recipes across the world."
- CulinaryDB: A comprehensive database of world recipes and ingredients.
- Dataset files used for analysis: 01_Recipe_Details.csv, 02_Ingredients.csv, 03_Compound_Ingredients.csv, 04_Recipe-Ingredients_Aliases.csv